Chapter 9, Problem 80P

Problem

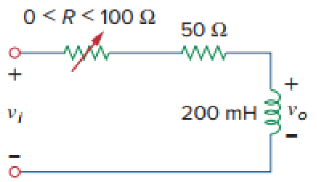
Consider the phase-shifting circuit in Fig. 9.83. Let **V***i* = 120 V operating at 60 Hz. Find:

(a) **V***o* when *R* is maximum

(b) **V***o* when *R* is minimum

(c) the value of *R* that will produce a phase shift of 45°

Figure 9.83:



Step-by-step solution

Step 1 of 4

$$L = 200\,\text{mH}$$

$$j\omega L = j(2\pi)(60)(200\times10^{-3})$$

$$j\omega L = j75.4\Omega$$

$$V_0 = \frac{j75.4}{R+50+j75.4}$$

$$V_1 = \frac{j75.4}{R+50+j75.4}(120\angle 0°)$$

Step 2 of 4

(a)

When R = 100 Ω

$$V_0 = \frac{j75.4}{150+j75.4}(120\angle 0°)$$

$$V_0 = \frac{(75.4\angle 90°)(120\angle 0°)}{167.88\angle 26.69°}$$

$$V_0 = 53.89\angle 63.31°\ \text{V}$$

Step 3 of 4

(b)

When R = 0 Ω

$$V_0 = \frac{j75.4}{50 + j75.4}(120\angle 0°)$$

$$V_0 = \frac{(75.4\angle 90°)(120\angle 0°)}{90.47\angle 56.45°}$$

$$V_0 = 100\angle 33.55° \text{ V}$$

Step 4 of 4

(c)

To produce a phase shift of $45°$.

The phase of $V_0 = 90° + 0° - \alpha = 45°$.

Hence,

$$\alpha = \text{Phase of} \quad (R + 50 + j75.4)$$

$$\alpha = 75°$$

For α to be $45°$

$$R + 50 = 75.4$$

Therefore,

$$R = 25.4\ \Omega$$